Describing Data:

Frequency Tables, Frequency Distributions, and Graphic Presentation

Chapter 02

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LEARNING OBJECTIVES

LO 2-1 Make a frequency table for a set of data.
LO 2-2 Organize data into a bar chart.
LO 2-3 Present a set of data using a pie chart.
LO 2-4 Create a frequency distribution for a data set.
LO 2-5 Understand a relative frequency distribution.
LO 2-6 Present data from a frequency distribution in a histogram or frequency polygon.
LO 2-7 Construct and interpret a cumulative frequency distribution.

LO 2-1 Make a frequency table for a set of data.

Frequency Table and Frequency Distribution

FREQUENCY TABLE A grouping of qualitative data into mutually exclusive classes showing the number of observations in each class.

Car Type	Number of Cars
Domestic	50
Foreign	30

Class interval: The class interval is obtained by subtracting the lower limit of a class from the lower limit of the next class.

Class frequency: The number of observations in each class.

Class midpoint: A point that divides a class into two equal parts. This is the average of the upper and lower class limits.

Location	Number of Cars
Kane	52
Olean	40
Sheffield	45
Tionesta	43
Total	180

FREQUENCY DISTRIBUTION A grouping of data into mutually exclusive classes showing the number of observations in each class.

LO 2-2 Organize data into a bar chart.

Bar Charts

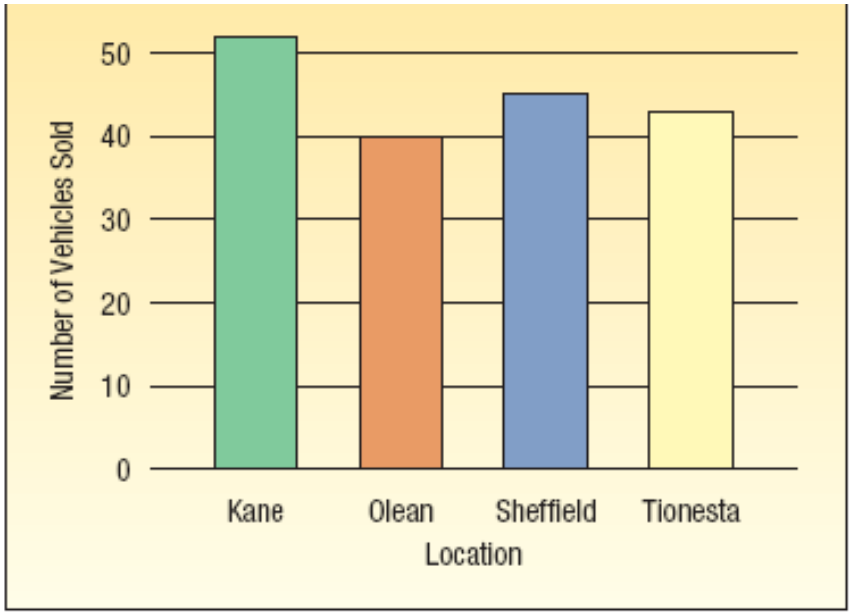


CHART 2–1 Number of Vehicles Sold by Location

BAR CHART A graph in which the classes are reported on the horizontal axis and the class frequencies on the vertical axis. The class frequencies are proportional to the heights of the bars.

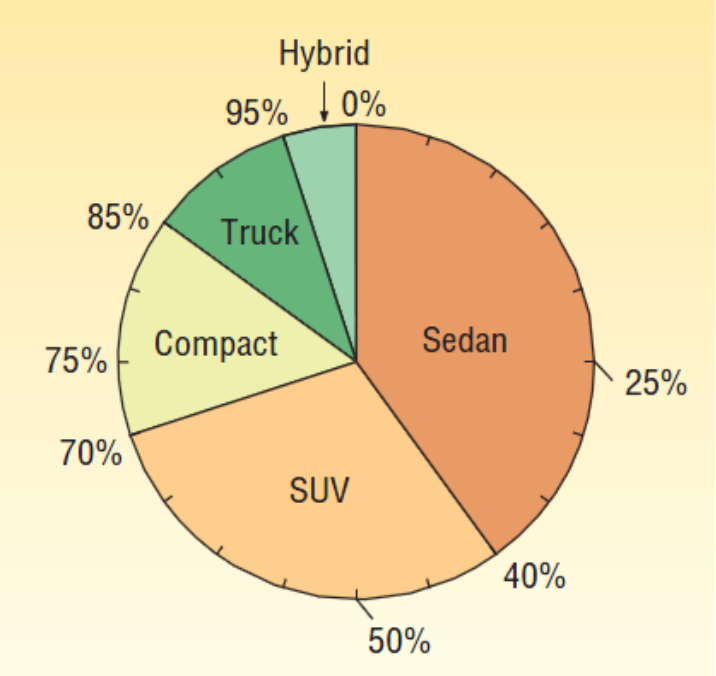
LO2-3 Organize data into a pie chart.

Pie Charts

PIE CHART A chart that shows the proportion or percent that each class represents of the total number of frequencies.

Vehicle Sales by Type at Applewood Auto Group

Vehicle Type	Number Sold	Percent Sold
Sedan	72	40
SUV	54	30
Compact	27	15
Truck	18	10
Hybrid	9	5
Total	180	100



LO 2-4 Create a frequency distribution for a data set.

EXAMPLE – Creating a Frequency Distribution Table

Ms. Kathryn Ball of AutoUSA wants to develop tables, charts, and graphs to show the typical selling price on various dealer lots. The table on the right reports only the price of the 80 vehicles sold last month at Whitner Autoplex.

TABLE 2–4 Profit on Vehicles Sold Last Month by the Applewood Auto Group

Highest

$1,387	$2,148	$2,201	$ 963	$ 820	$2,230	$3,043	$2,584	$2,370
1,754	2,207	996	1,298	1,266	2,341	1,059	2,666	2,637
1,817	2,252	2,813	1,410	1,741	3,292	1,674	2,991	1,426
1,040	1,428	323	1,553	1,772	1,108	1,807	934	2,944
1,273	1,889	352	1,648	1,932	1,295	2,056	2,063	2,147
1,529	1,166	482	2,071	2,350	1,344	2,236	2,083	1,973
3,082	1,320	1,144	2,116	2,422	1,906	2,928	2,856	2,502
1,951	2,265	1,485	1,500	2,446	1,952	1,269	2,989	783
2,692	1,323	1,509	1,549	369	2,070	1,717	910	1,538
1,206	1,761	1,638	2,348	978	2,454	1,797	1,536	2,339
1,342	1,919	1,961	2,498	1,238	1,606	1,955	1,957	2,700
443	2,357	2,127	294	1,818	1,680	2,199	2,240	2,222
754	2,866	2,430	1,115	1,824	1,827	2,482	2,695	2,597
1,621	732	1,704	1,124	1,907	1,915	2,701	1,325	2,742
870	1,464	1,876	1,532	1,938	2,084	3,210	2,250	1,837
1,174	1,626	2,010	1,688	1,940	2,639	377	2,279	2,842
1,412	1,761	2,165	1,822	2,197	842	1,220	2,626	2,434
1,809	1,915	2,231	1,897	2,646	1,963	1,401	1,501	1,640
2,415	2,119	2,389	2,445	1,461	2,059	2,175	1,752	1,821
1,546	1,766	335	2,886	1,731	2,338	1,118	2,058	2,487

Lowest

Constructing a Frequency Table - Example

- **Step 1: Decide on the number of classes.**

  A useful recipe to determine the number of classes (k) is the "2 to the k rule." such that $2^k > n$.

  There were 180 vehicles sold, so $n = 180$. If we try $k = 7$, then $2^7 = 128$, somewhat less than 180. Hence, 7 is not enough classes. If we let $k = 8$, then $2^8 = 256$, which is greater than 180. So the recommended number of classes is 8.

- **Step 2: Determine the class interval or width.**

  The formula is: $i \geq (H - L)/k$ where i is the class interval, H is the highest observed value, L is the lowest observed value, and k is the number of classes.

$$i \geq \frac{H - L}{k} = \frac{\$3{,}292 - \$294}{8} = \$374.75$$

  Round up to some convenient number, such as a multiple of 10 or 100. Use a class width of \$400.

Constructing a Frequency Table - Example

- **Step 3: Set the individual class limits.**

Classes
$ 200 up to $ 600
600 up to 1,000
1,000 up to 1,400
1,400 up to 1,800
1,800 up to 2,200
2,200 up to 2,600
2,600 up to 3,000
3,000 up to 3,400

- **Step 4: Tally the vehicle selling prices into the classes.**

TABLE 2–7 Frequency Distribution of Profit for Vehicles Sold Last Month at Applewood Auto Group

Profit	Frequency
$ 200 up to $ 600	卌 III
600 up to 1,000	卌 卌 I
1,000 up to 1,400	卌 卌 卌 卌 III
1,400 up to 1,800	卌 卌 卌 卌 卌 卌 卌 III
1,800 up to 2,200	卌 卌 卌 卌 卌 卌 卌 卌 卌
2,200 up to 2,600	卌 卌 卌 卌 卌 卌 II
2,600 up to 3,000	卌 卌 卌 IIII
3,000 up to 3,400	IIII
Total	

- **Step 5: Count the number of items in each class.**

Profit	Frequency
$ 200 up to $ 600	8
600 up to 1,000	11
1,000 up to 1,400	23
1,400 up to 1,800	38
1,800 up to 2,200	45
2,200 up to 2,600	32
2,600 up to 3,000	19
3,000 up to 3,400	4
Total	180

LO 2-5 Understand a relative frequency distribution.

Relative Class Frequencies

- Class frequencies can be converted to **relative class frequencies** to show the fraction of the total number of observations in each class.
- A relative frequency captures the relationship between a class total and the total number of observations.

TABLE 2–2 Relative Frequency Table of Vehicles Sold By Type At Whitner Autoplex Last Month

Vehicle Type	Number Sold	Relative Frequency
Domestic	50	0.625
Foreign	30	0.375
Total	80	1.000

Relative Frequency Distribution

To convert a frequency distribution to a *relative* frequency distribution, each of the class frequencies is divided by the total number of observations.

TABLE 2–8 Relative Frequency Distribution of Profit for Vehicles Sold Last Month at Applewood Auto Group

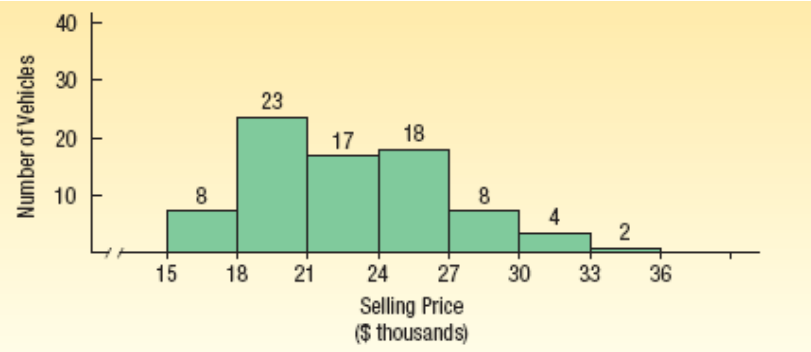
Profit	Frequency	Relative Frequency	Found by
$ 200 up to $ 600	8	.044	8/180
600 up to 1,000	11	.061	11/180
1,000 up to 1,400	23	.128	23/180
1,400 up to 1,800	38	.211	38/180
1,800 up to 2,200	45	.250	45/180
2,200 up to 2,600	32	.178	32/180
2,600 up to 3,000	19	.106	19/180
3,000 up to 3,400	4	.022	4/180
Total	180	1.000	

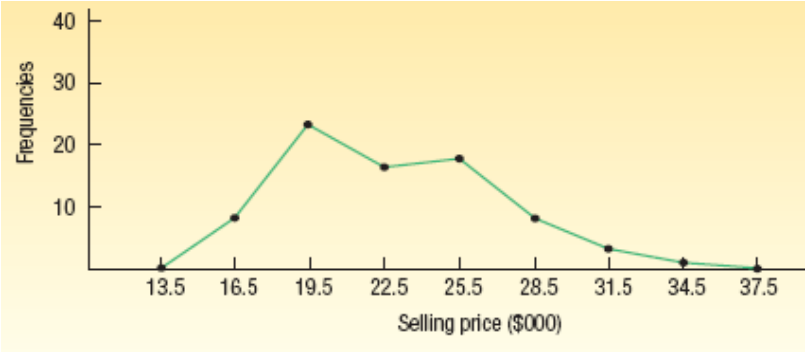
LO 2-6 Present data from a frequency distribution in a histogram or frequency polygon.

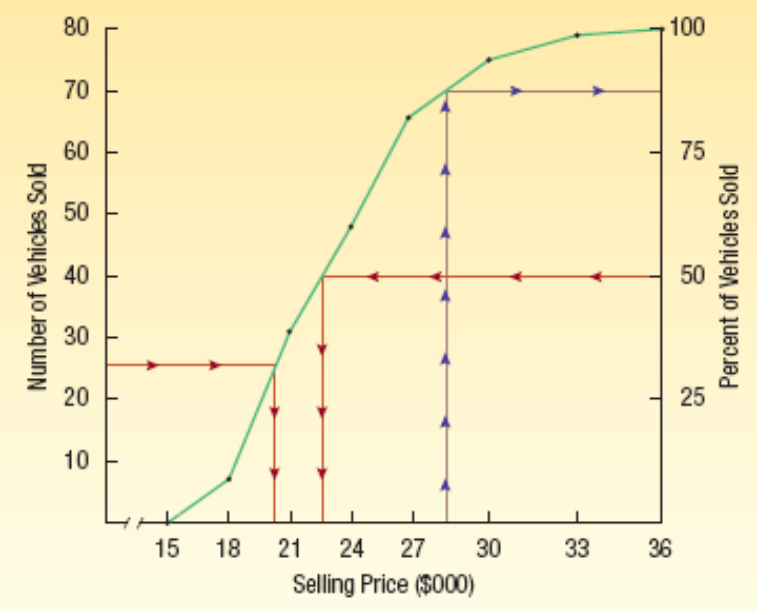
Graphic Presentation of a Frequency Distribution

The three commonly used graphic forms are:

- Histograms
- Frequency polygons
- Cumulative frequency distributions







Histogram

HISTOGRAM A graph in which the classes are marked on the horizontal axis and the class frequencies on the vertical axis. The class frequencies are represented by the heights of the bars and the bars are drawn adjacent to each other.

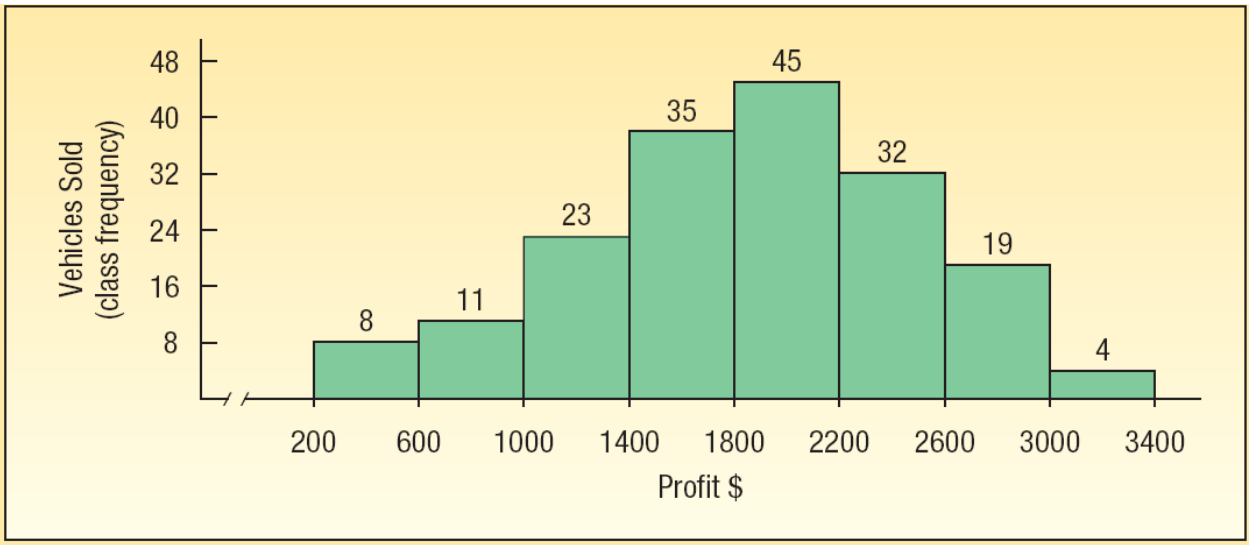


CHART 2–4 Histogram of the Profit on 180 Vehicles Sold at the Applewood Auto Group

LO 2-6

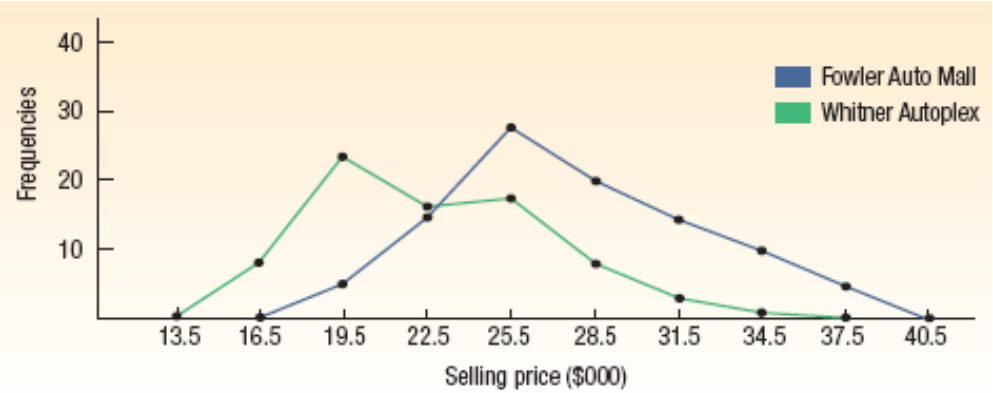
Frequency Polygon

- A **frequency polygon** also shows the shape of a distribution and is similar to a histogram.
- It consists of line segments connecting the points formed by the intersections of the class midpoints and the class frequencies.

Profit	Midpoint	Frequency
$ 200 up to $ 600	$ 400	8
600 up to 1,000	800	11
1,000 up to 1,400	1,200	23
1,400 up to 1,800	1,600	38
1,800 up to 2,200	2,000	45
2,200 up to 2,600	2,400	32
2,600 up to 3,000	2,800	19
3,000 up to 3,400	3,200	4
Total		180

Histogram Versus Frequency Polygon

- Both provide a quick picture of the main characteristics of the data (highs, lows, points of concentration, etc.).
- The histogram has the advantage of depicting each class as a rectangle, with the height of the rectangular bar representing the number in each class.
- The frequency polygon has an advantage over the histogram. It allows us to compare directly two or more frequency distributions.



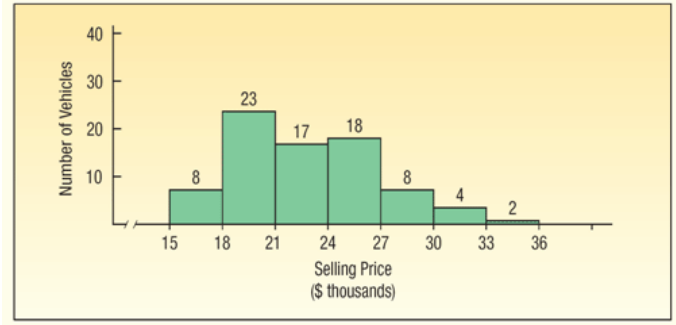


CHART 2–4 Histogram of the Selling Prices of 80 Vehicles at Whitner Autoplex

LO 2-7 Construct and interpret a cumulative frequency distribution.

Cumulative Frequency Distribution

TABLE 2–9 Cumulative Frequency Distribution for Profit on Vehicles Sold Last Month at Applewood Auto Group

Profit	Frequency	Cumulative Frequency	Found by
$ 200 up to $ 600	8	8	8
600 up to 1,000	11	19	8 + 11
1,000 up to 1,400	23	42	8 + 11 + 23
1,400 up to 1,800	38	80	8 + 11 + 23 + 30
1,800 up to 2,200	45	125	8 + 11 + 23 + 30 + 45
2,200 up to 2,600	32	157	8 + 11 + 23 + 30 + 45 + 32
2,600 up to 3,000	19	176	8 + 11 + 23 + 30 + 45 + 32 + 19
3,000 up to 3,400	4	180	8 + 11 + 23 + 30 + 45 + 32 + 19 + 4
Total	180		

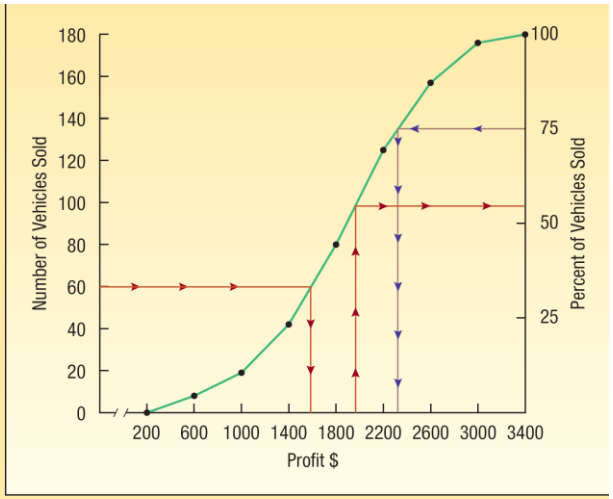


CHART 2–7 Cumulative Frequency Distribution for Vehicle Profit